Week 4 Home Work

Submission Deadline: 16 May 2026, 11:59 PM

1. Voting Eligibility Checker

Problem Statement:

- Take the user's age as input.
- Convert the input into integer.
- Check whether age is 18 or above.
- Print proper eligibility message.

Acceptance Criteria:

- Age >= 18 → Print: Eligible for voting
- Age < 18 → Print: Not eligible for voting

Expected Output:

```
Enter your age: 20
Eligible for voting
```

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2. Student Grade System

Problem Statement:

- Take student's marks as input.
- Use if-elif-else to determine grade.
- Print the grade based on marks.

Acceptance Criteria:

- Marks >= 90 → A+
- Marks >= 80 → A
- Marks >= 70 → B
- Marks >= 60 → C
- Marks < 60 → F (Fail)

Expected Output:

```
Enter your marks: 85
A
```

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3. Restaurant Membership Discount

Problem Statement:

- Take membership status (yes/no).
- Take total bill amount.
- Members get 10% discount.
- Non-members pay full price.
- Print final bill.

Acceptance Criteria:

- Member (yes): Final bill = bill amount × 0.90
- Non-member (no): Final bill = full bill amount

Expected Output:

```
Are you a member? yes
Enter bill amount: 5000
Final bill: 4500.0
```

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4. Username Validator

Problem Statement:

- Take username input.
- Username must contain only alphabets.
- Length must be at least 5 characters.
- Print validity message.

Acceptance Criteria:

- Only alphabets AND length >= 5 → Valid username
- Contains non-alphabets OR length < 5 → Invalid username

Expected Output:

```
Enter username: Refadul
Valid username
```

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5. ATM PIN Verification

Problem Statement:

- Take PIN input.
- Verify entered PIN with the correct PIN.
- Print access message.

Acceptance Criteria:

- Correct PIN is: 1234
- Entered PIN == 1234 → Access Granted
- Entered PIN != 1234 → Access Denied

Expected Output:

```
Enter PIN: 1234
Access Granted
```

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6. Electricity Bill Warning

Problem Statement:

- Take electricity unit input.
- Check whether usage exceeds 500 units.
- Print warning accordingly.

Acceptance Criteria:

- Units > 500 → Print: High Bill
- Units <= 500 → Print: Normal Bill

Expected Output:

```
Enter units: 700
High Bill
```

7. Password Strength Checker

Problem Statement:

- Take password input.
- Password must be at least 8 characters long.
- Password must contain at least one digit.
- Print strength message.

Acceptance Criteria:

- Length >= 8 AND contains at least one digit → Strong Password
- Otherwise → Weak Password

Expected Output:

```
Enter password: hello123
Strong Password
```

8. Online Course Access

Problem Statement:

- Take age and country input.
- Access allowed if age >= 18 OR country is Japan.
- Print access message.

Acceptance Criteria:

- Age >= 18 OR country == 'Japan' → Access Granted
- Age < 18 AND country != 'Japan' → Access Denied

Expected Output:

```
Enter age: 17
Enter country: Japan
Access Granted
```

9. Product Stock Checker

Problem Statement:

- Take product quantity input.
- Check whether quantity is greater than zero.
- Print stock status.

Acceptance Criteria:

- Quantity > 0 → In Stock
- Quantity <= 0 → Out of Stock

Expected Output:

```
Enter quantity: 5
In Stock
```

10. Email Validator

Problem Statement:

- Take email input.

- Email must contain '@'.
- Email must end with '.com'.
- Print validity message.

Acceptance Criteria:

- Contains '@' AND ends with '.com' → Valid Email
- Missing '@' OR does not end with '.com' → Invalid Email

Expected Output:

```
Enter email: abc@gmail.com
Valid Email
```

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11. Bus Fare System

Problem Statement:

- Take passenger age.
- Calculate fare based on age category.
- Print the fare in yen.

Acceptance Criteria:

- Age <= 12 (Child) → Fare: 150 yen
- Age <= 17 (Teen) → Fare: 300 yen
- Age <= 59 (Adult) → Fare: 500 yen
- Age >= 60 (Senior) → Fare: 200 yen

Expected Output:

```
Enter age: 15
Bus fare: 300 yen
```

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12. Movie Ticket Eligibility

Problem Statement:

- Take age input.
- Only users aged 18 or above can watch the movie.
- Print eligibility message.

Acceptance Criteria:

- Age >= 18 → You can watch the movie
- Age < 18 → You cannot watch the movie

Expected Output:

```
Enter age: 22
You can watch the movie
```

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13. Login System with Nested Conditions

Problem Statement:

- Take username and password.
- Verify username first, then verify password.
- Print login result.

Acceptance Criteria:

- Correct username: admin
- Correct password: 12345
- Both correct → Login Successful
- Wrong username → Invalid Username
- Wrong password → Wrong Password

Expected Output:

```
Enter username: admin
Enter password: 12345
Login Successful
```

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14. CGPA Scholarship Checker

Problem Statement:

- Take CGPA input.
- Determine scholarship category based on CGPA.
- Print scholarship result.

Acceptance Criteria:

- CGPA >= 3.80 → Full Scholarship
- CGPA >= 3.50 → Half Scholarship
- CGPA >= 3.00 → Quarter Scholarship
- CGPA < 3.00 → No Scholarship

Expected Output:

```
Enter CGPA: 3.85
Full Scholarship
```

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15. Mobile Recharge Bonus

Problem Statement:

- Take recharge amount.
- Provide bonus for recharge above 1000 yen.
- Print bonus result.

Acceptance Criteria:

- Amount > 1000 → Bonus Added: 100 yen
- Amount <= 1000 → No Bonus

Expected Output:

```
Enter recharge amount: 1500
Bonus Added: 100 yen
```

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16. Delivery Charge Calculator

Problem Statement:

- Take shopping amount.
- Orders above 5000 yen get free delivery.
- Otherwise, a delivery charge is applied.

- Print delivery result.

Acceptance Criteria:

- Amount > 5000 → Free Delivery
- Amount <= 5000 → Delivery Charge: 500 yen

Expected Output:

```
Enter amount: 7000
Free Delivery
```

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17. Website Age Restriction

Problem Statement:

- Take user's age.
- Users below 13 cannot create accounts.
- Print account creation result.

Acceptance Criteria:

- Age >= 13 → Account creation allowed
- Age < 13 → Account creation denied

Expected Output:

```
Enter age: 12
Account creation denied
```

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18. Salary Tax Calculator

Problem Statement:

- Take salary input.
- Calculate tax percentage based on salary range.
- Print the tax amount.

Acceptance Criteria:

- Salary > 1,000,000 → Tax rate: 30%
- Salary > 500,000 → Tax rate: 10%
- Salary <= 500,000 → Tax rate: 0% (No tax)

Expected Output:

```
Enter salary: 600000
Tax: 60000.0
```

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19. Traffic Signal System

Problem Statement:

- Take traffic signal color.
- Print appropriate action.

Acceptance Criteria:

- green → Go
- yellow → Slow Down
- red → Stop

- Other → Invalid Signal

Expected Output:

```
Enter signal: green
Go
```

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20. Student ID Checker

Problem Statement:

- Take student ID input.
- Check whether ID starts with '260'.
- Print validity message.

Acceptance Criteria:

- ID starts with '260' → Valid Student
- ID does not start with '260' → Invalid Student

Expected Output:

```
Enter student ID: 2600240464
Valid Student
```

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21. Advanced E-commerce Discount System

Problem Statement:

- Take membership status and cart amount.
- Apply discount rules based on amount range.
- Print final payable bill.

Acceptance Criteria:

- Member + Amount >= 20,000 → 3% discount
- Member + Amount >= 10,000 → 2% discount
- Member + Amount < 10,000 → 1% discount
- Non-member → No discount (full price)

Expected Output:

```
Are you a member? yes
Enter amount: 30000
Final bill: 29100.0
```

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22. Smart Login Security System

Problem Statement:

- Take username, password, and OTP.
- Verify all credentials before login approval.
- Print login result.

Acceptance Criteria:

- Correct username: admin
- Correct password: 12345
- Correct OTP: 9999

- All three correct → Login Approved
- Wrong username → Invalid Username
- Wrong password → Wrong Password
- Wrong OTP → Invalid OTP

Expected Output:

```
Enter username: admin
Enter password: 12345
Enter OTP: 9999
Login Approved
```

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23. University Admission System

Problem Statement:

- Take GPA and IELTS score.
- Check minimum eligibility requirements.
- Print admission result.

Acceptance Criteria:

- GPA >= 3.5 AND IELTS >= 6.5 → Eligible for admission
- GPA < 3.5 → GPA requirement not met
- IELTS < 6.5 → IELTS requirement not met

Expected Output:

```
Enter GPA: 3.8
Enter IELTS: 7
Eligible for admission
```

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24. Bank Loan Eligibility System

Problem Statement:

- Take salary and years of experience.
- Approve loan if both conditions are satisfied.
- Print loan result.

Acceptance Criteria:

- Salary >= 300,000 AND Experience >= 2 years → Loan Approved
- Salary < 300,000 → Salary requirement not met
- Experience < 2 → Experience requirement not met

Expected Output:

```
Enter salary: 400000
Enter experience: 3
Loan Approved
```

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25. Hospital Emergency Priority System

Problem Statement:

- Take age and emergency level (scale 1-10).
- Determine patient treatment priority.

- Print priority level.

Acceptance Criteria:

- Age >= 60 OR emergency level >= 7 → Priority Treatment
- Age >= 18 AND emergency level >= 4 → Normal Treatment
- Otherwise → Standard Queue

Expected Output:

```
Enter age: 65
Enter emergency level: 6
Priority Treatment
```

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26. Flight Ticket System

Problem Statement:

- Take nationality and passport validity (yes/no).
- Allow booking only if passport is valid.
- Print booking result.

Acceptance Criteria:

- Passport valid (yes) → Ticket Booking Allowed
- Passport invalid (no) → Ticket Booking Denied

Expected Output:

```
Enter nationality: Bangladesh
Is passport valid? yes
Ticket Booking Allowed
```

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27. Smart Attendance System

Problem Statement:

- Take attendance percentage and assignment submission status (yes/no).
- Students need both conditions fulfilled for exam eligibility.
- Print eligibility result.

Acceptance Criteria:

- Attendance >= 75% AND assignment submitted (yes) → Eligible for final exam
- Attendance < 75% → Not eligible (low attendance)
- Assignment not submitted (no) → Not eligible (missing assignment)

Expected Output:

```
Enter attendance: 80
Assignment submitted? yes
Eligible for final exam
```

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28. AI Course Enrollment System

Problem Statement:

- Take skill level (beginner / intermediate / advanced) and age.
- Approve enrollment according to conditions.
- Print enrollment result.

Acceptance Criteria:

- Skill: beginner AND age >= 18 → Enrollment Approved
- Skill: intermediate AND age >= 16 → Enrollment Approved
- Skill: advanced AND age >= 14 → Enrollment Approved
- Age below required threshold → Enrollment Denied
- Invalid skill level → Invalid Input

Expected Output:

```
Enter skill level: beginner
Enter age: 20
Enrollment Approved
```

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29. Smart Parking System

Problem Statement:

- Take vehicle type (bike / car / truck) and parking hours.
- Calculate total parking fee based on vehicle type and duration.
- Print total fee.

Acceptance Criteria:

- bike → 100 yen/hour
- car → 300 yen/hour
- truck → 500 yen/hour
- Other → Invalid vehicle type

Expected Output:

```
Enter vehicle type: car
Enter parking hours: 5
Total fee: 1500 yen
```

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30. Smart Software Engineer Recruitment System

Problem Statement:

- Take Python skill score, problem solving score, communication skill (good/bad), and years of experience.
- Check all hiring conditions.
- Print recruitment result.

Acceptance Criteria:

- Python skill score >= 80
- Problem solving score >= 75
- Communication skill == good
- Years of experience >= 2
- All 4 conditions met → Selected for Final HR Round
- Any condition fails → Not Selected

Expected Output:

```
Enter Python skill score: 85
Enter problem solving score: 80
```

```
Enter communication skill (good/bad): good
Enter years of experience: 2
Selected for Final HR Round
```